ELECTRIC VEHICLES MARKET PERFORMANCE

Project Documentation

Submitted by

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Introduction

Electric vehicles (EVs) are becoming increasingly popular as an alternative to conventional fuel-powered vehicles. They help reduce dependence on petrol and diesel and support cleaner transportation.

This project, “Electric Vehicles Market Performance,” uses Tableau to analyse electric vehicle market data and understand its major trends and performance through data visualization.

Problem Statement

The electric vehicle market is growing rapidly, and analysing its performance helps us understand market trends. This project uses data visualization to study electric vehicle data and present useful insights through Tableau charts, dashboards, and stories.

Objectives

The main objectives of this project are:

- To analyse electric vehicle market data.
- To identify important trends and patterns.
- To create meaningful visualizations using Tableau.
- To develop a dashboard and a story.
- To present the key findings of the analysis.

Data Collection

The dataset used in this project contains information related to the electric vehicle market. It was collected for analysing electric vehicle performance and identifying market trends.

The dataset was imported into Tableau to explore the data and create visualizations, dashboards, and a story for presenting the results.

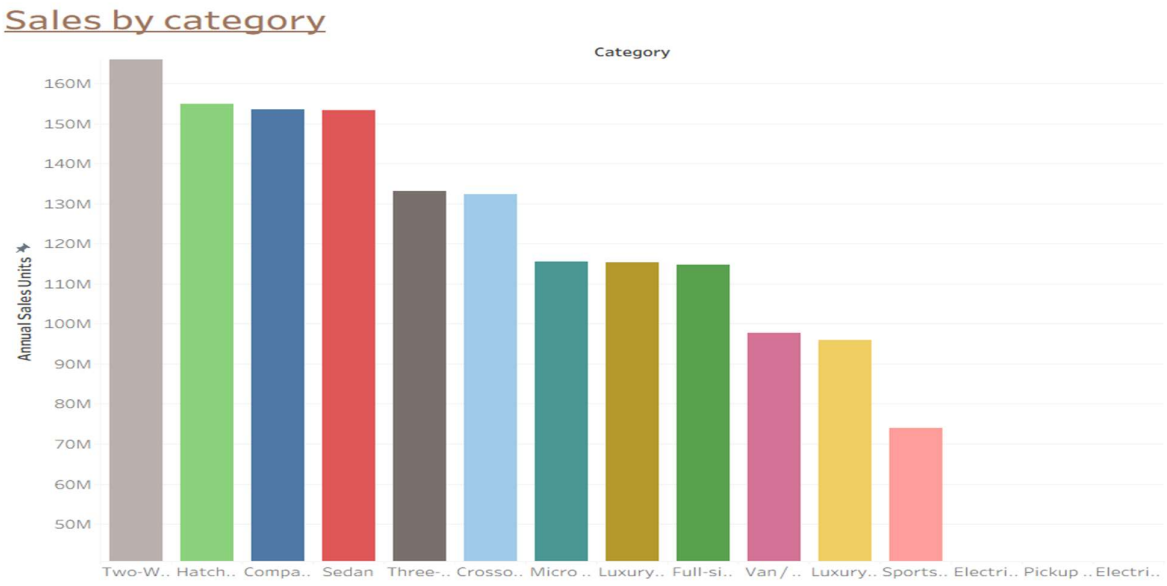
Data Cleaning and Preprocessing

Data cleaning and preprocessing are important steps in data analysis. The dataset was checked to understand its structure and prepare it for visualization. The data was organized into a suitable format so that it could be analysed effectively in Tableau.

After preparing the data, different charts were created to understand electric vehicle market trends and performance.

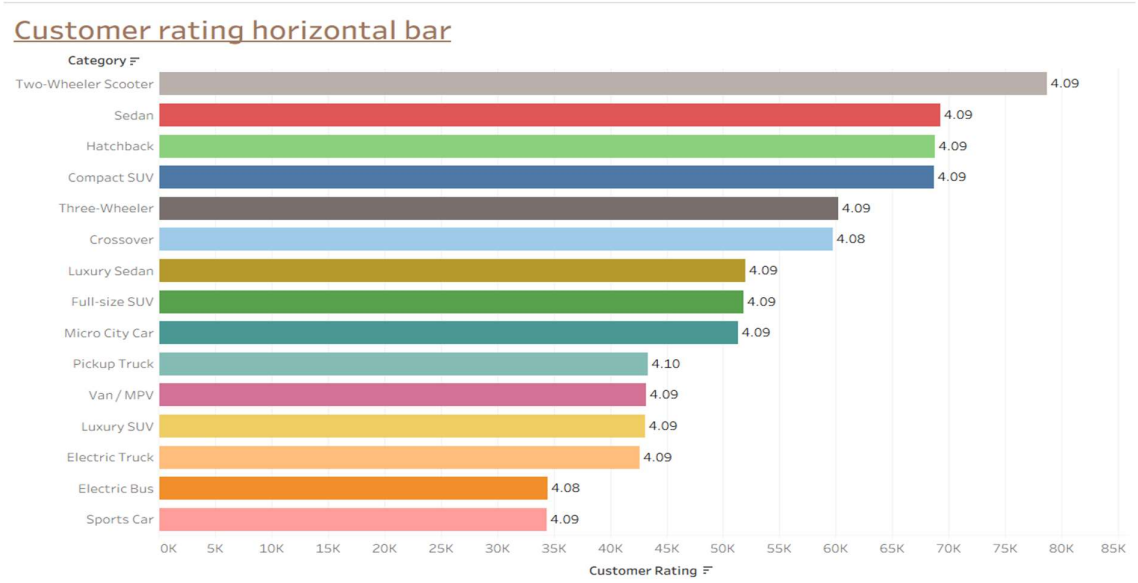
Data Visualization

In this project, Tableau was used to create six visualizations to analyse electric vehicle market performance. These charts help present the data clearly and identify important patterns and trends.



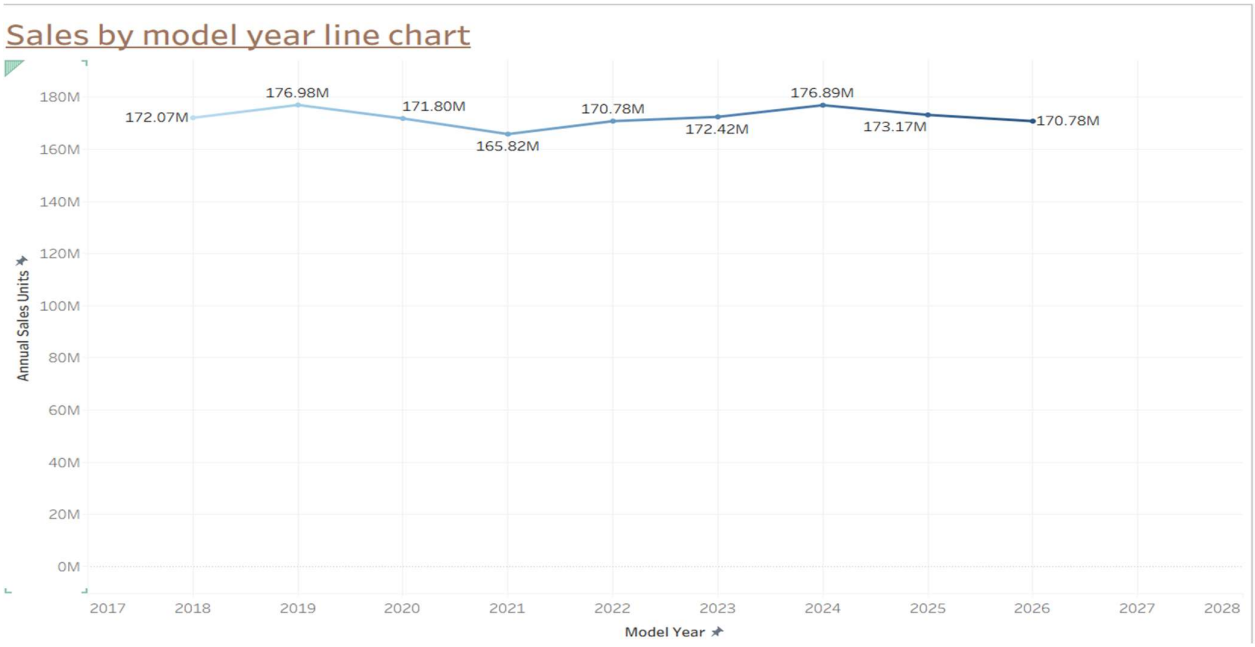
1. Sales by Category

This bar chart compares sales across different categories. It helps identify which categories contribute more to overall sales and makes category-wise performance easy to understand.



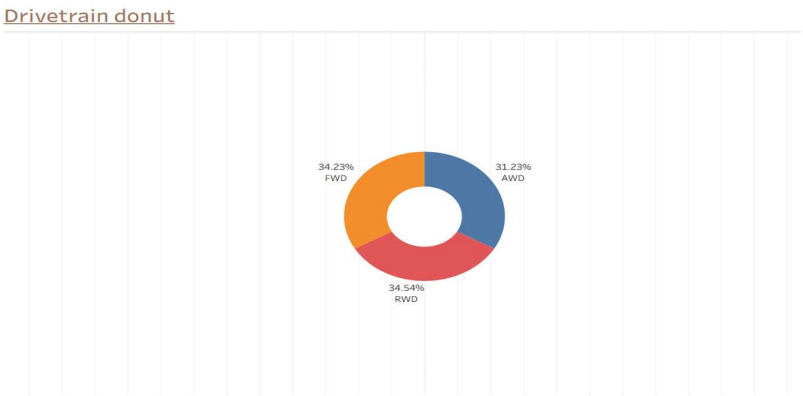
2. Customer Rating

This horizontal bar chart displays customer ratings across different categories or groups. It helps compare customer satisfaction levels and identify which categories receive higher ratings.



3. Sales by Model Year

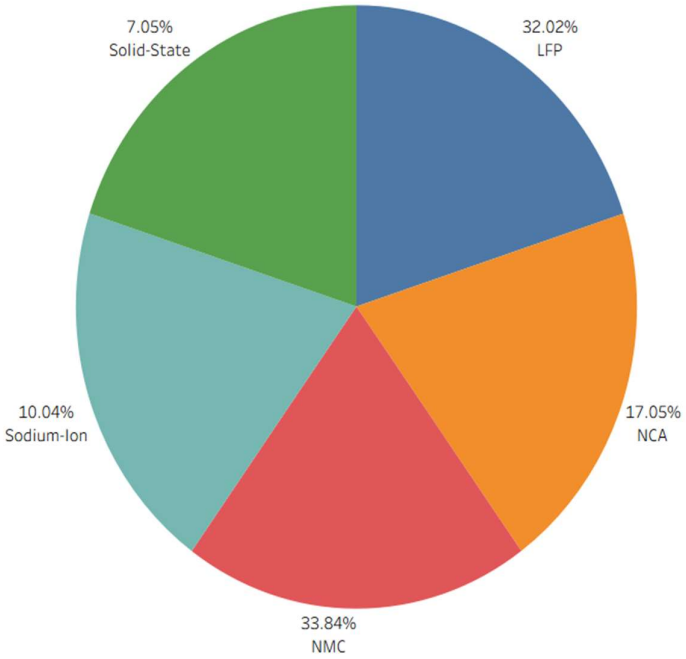
This line chart shows the trend in sales across different vehicle model years. It helps identify changes in sales over time and highlights the model years with higher or lower sales.



4. Drivetrain Distribution

This donut chart shows the distribution of electric vehicles by drivetrain type. It helps compare the proportion of vehicles using different drivetrain configurations and understand their share in the dataset.

Battery type pie chart



5. Battery Type Distribution

This pie chart shows the distribution of electric vehicles by battery type. It helps compare the share of different battery types in the dataset and understand which types are more common.

Market region Heatmap chart

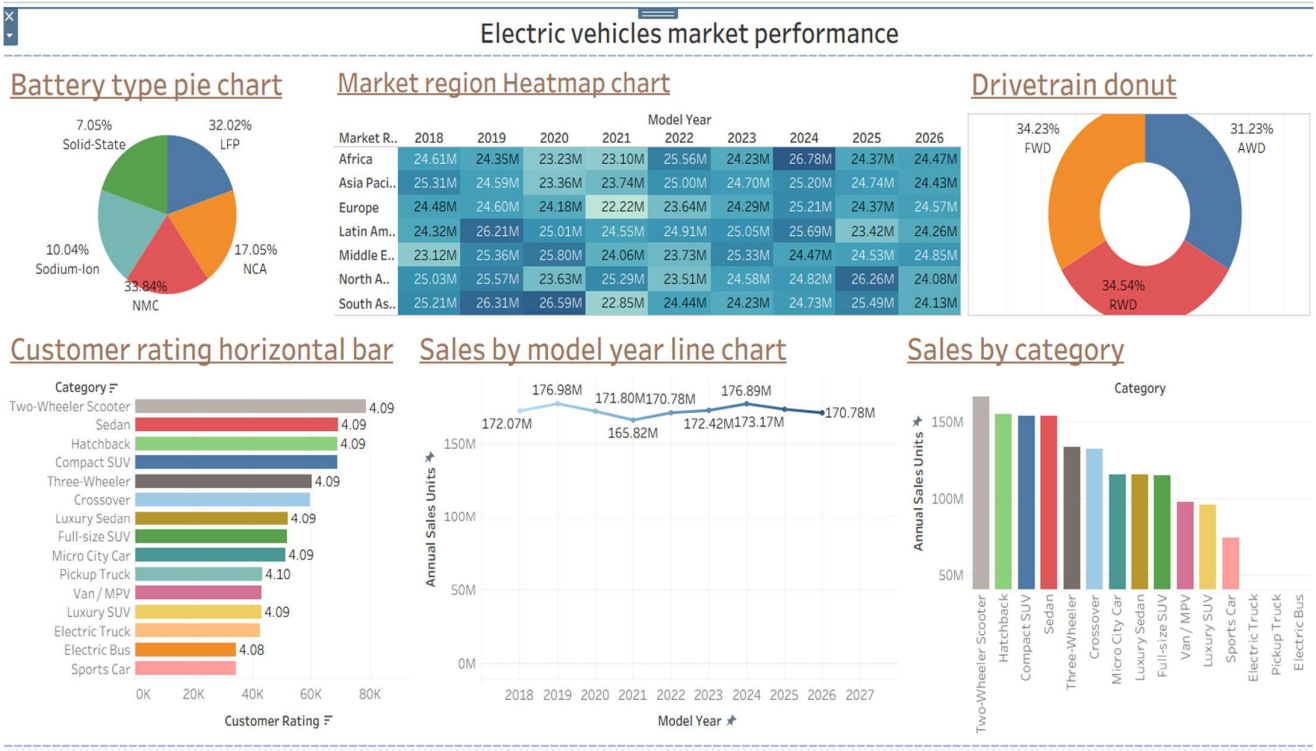
Market Region	Model Year 2018	2019	2020	2021	2022	2023	2024	2025	2026
Africa	24.61M	24.35M	23.23M	23.10M	25.56M	24.23M	26.78M	24.37M	24.47M
Asia Pacific	25.31M	24.59M	23.36M	23.74M	25.00M	24.70M	25.20M	24.74M	24.43M
Europe	24.48M	24.60M	24.18M	22.22M	23.64M	24.29M	25.21M	24.37M	24.57M
Latin America	24.32M	26.21M	25.01M	24.55M	24.91M	25.05M	25.69M	23.42M	24.26M
Middle East	23.12M	25.36M	25.80M	24.06M	23.73M	25.33M	24.47M	24.53M	24.85M
North America	25.03M	25.57M	23.63M	25.29M	23.51M	24.58M	24.82M	26.26M	24.08M
South Asia	25.21M	26.31M	26.59M	22.85M	24.44M	24.23M	24.73M	25.49M	24.13M

6. Market Region Heatmap

This heatmap presents electric vehicle market performance across different regions. The colour variations help compare regional values and identify areas with relatively higher or lower market activity.

Tableau Dashboard

A dashboard was created in Tableau by combining the visualizations developed for the Electric Vehicles Market Performance project. It presents important information in one place and helps users compare different aspects of the electric vehicle market easily.



Key Findings

The analysis of the electric vehicle market provided the following insights:

- The Sales by Category chart helps compare sales across categories.
- The Customer Rating chart shows differences in customer ratings.
- The Sales by Model Year chart highlights sales trends over time.
- The Drivetrain Distribution chart compares drivetrain types.
- The Battery Type chart shows the distribution of battery types.
- The Market Region Heatmap helps compare market performance across regions.

Conclusion

This project focused on analysing Electric Vehicles Market Performance using Tableau. Six visualizations were created to explore

sales by category, customer ratings, model year, drivetrain, battery type, and market region. The dashboard brings important information together and makes the results easier to understand. This project helped develop practical skills in data visualization and data analysis.

References

1. Dataset used for the Electric Vehicles Market Performance project.

2. Tableau — Data visualization and dashboard development software. https://www.tableau.coms